Data availability

Chemical
pH (1:5)
E.C. (mS/cm)
Macronutrients (N,Ca, P...)
Micronutrients (Fe,Mb,Cu...)
Active Calcium
Carbonates (% CO_3)
CEC (meq/100g)
Heavy metals

Biological
Organic Matter (%)
C/N Ratio
Organic carbon (%)

Physical
Clays (%)
Silts (%)
Sands (%)
Texture



Risks

Climate variability, particularly prolonged droughts and heavy rainfall events that complicate cover crop management and soil carbon stability. There is also a risk of administrative burden for farmers, derived from certification and control requirements. Economically, fluctuations in olive oil prices can dilute the added value of IP if consumer communication is not strengthened.

Preliminary results

The system offers clear results: greater soil resilience against erosion and degradation, increases in soil organic carbon in surface horizons, reduced pest pressure through integrated pest management, and improved market image of the product. In Mediterranean soils analysed under the NOVASOIL project, the following outcomes were observed:

- Measurable **increases in soil organic carbon** and structural stability, mainly in the alleys.
- **Reduction of erosion** in sloping areas through well-managed cover crops.
- Greater **efficiency in the use of inputs**, with reduced mineral nitrogen applications.
- **Certified products** (olives and olive oil) with recognised environmental and social value.

Economic perspective

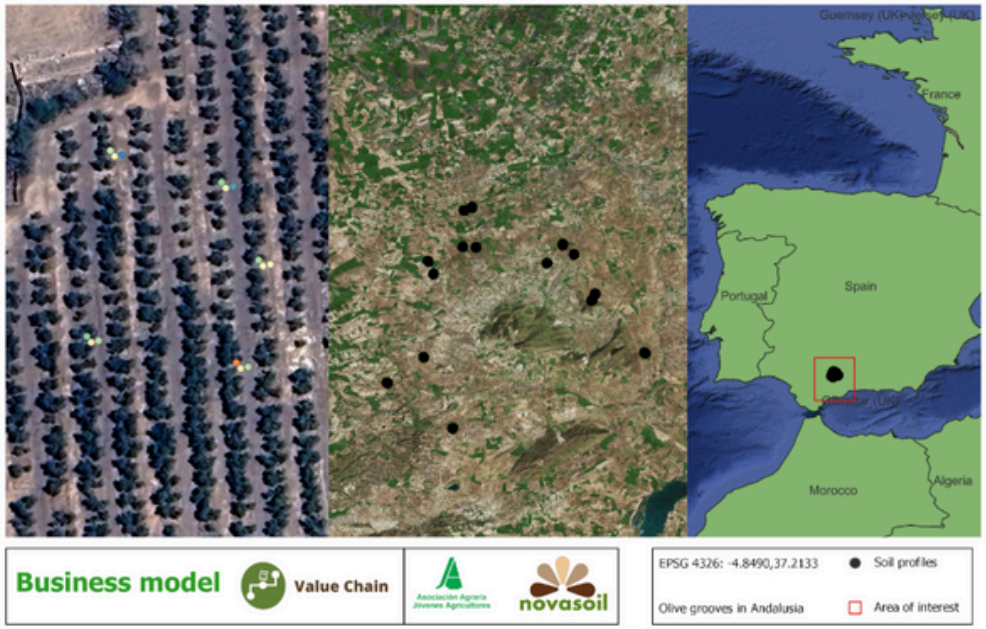
The profitability of the model is built on three pillars:

- **Controlled costs**, thanks to reduced use of external inputs and the reuse of local biomass (shredded prunings, spontaneous covers).
- **Diversified income**, through differentiated sales with the IP label, public support, and, in the future, possible market premiums linked to carbon footprint reduction or the provision of ecosystem services.
- Cooperative scale, which facilitates the aggregation of **certified product**, **reduces audit costs**, and consolidates the brand in international markets.

Policy and governance

The Junta de Andalucía, through its Integrated Production system, defines the technical crop standards, provides updated information on phytosanitary risks, and controls compliance through authorised bodies. The system relies on a community of stakeholders: farmers, cooperatives, mills, technical advisory companies, and certification bodies, all connected with the market.

Olives in integrated production



novasoil

The NOVASOIL project aims to show the benefits for society and the environment of investing in soil health.

The main output of the project will be a tool to analyse the suitability of different business cases that promote soil health.

This tool will be based on best practice examples in Europe and other countries, as well as on the needs and demands of society.

Integrated Production



Summarize

The integrated olive grove is a successful business model that promotes soil health and sustainability. It is a reliable and profitable way to produce both table olives and olive oil. The regional government provides information on modifications and pest risk forecasts to ensure the product meets environmental and social quality standards.

The NOVASOIL project aims to analyze the feasibility and characteristics of this model and replicate it in other areas or crops. The project involves a stakeholder community that includes the main players in the sector and their links to the market. The study includes 20 commercial plots with representative soils of the Mediterranean area to provide necessary information for the project's objectives

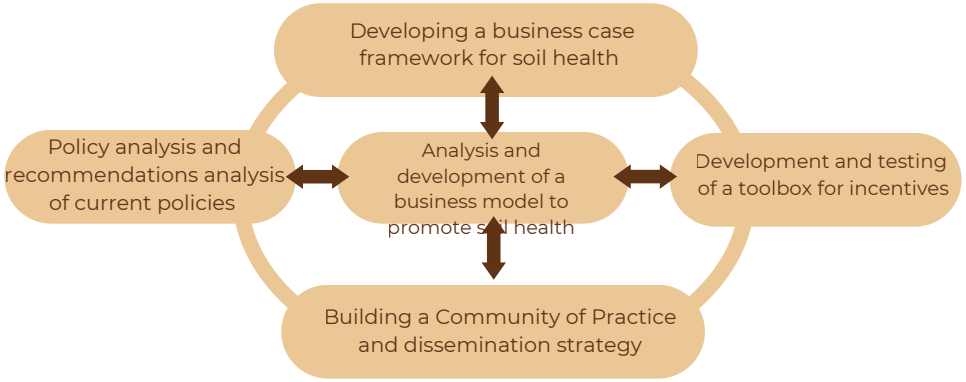


Figure 1. Structure of the NOVASOIL project

Case studies properties

Potential inhibitors

Economic crisis, Climate Change, competition with products from other countries with a lower cost.

Potential enablers

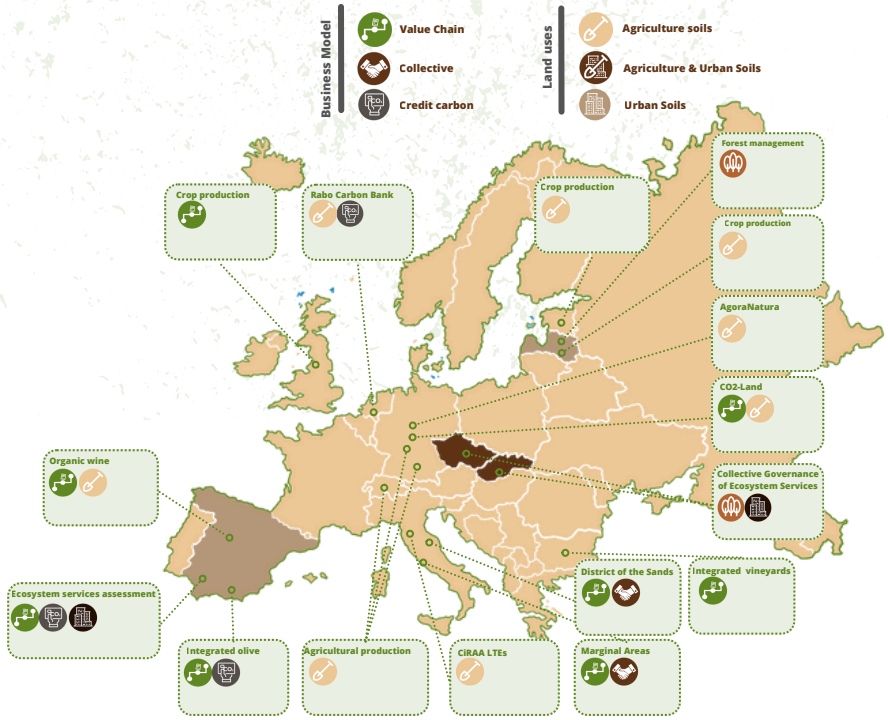
Social awareness (interested in high quality and healthy product)
Existing farmer network and policies.

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Potential stakeholders

APICOLIVAR, ASECOL, OLEOESTEPA, OPRACOL, Regional administration, Farmer association, Oil press.

.



Ecosystem services



Climate regulation – carbon storage

Soil quality

Specific objectives

SO1 Multi-dimensional portfolio development of business cases and good examples for investing in soil health in Europe and beyond.

SO2 Analysis of current models around the world for creating new incentives and new revenues from healthy soils.

SO3 Apply a co-development approach to modelling that promotes soil health.

SO4 Develop a prototype incentive toolbox and test network.

SO5 Analysis of current policies and technologies related to soil health and income provision for farmers.

SO6 Developing a community of practice around investors and key soil health actors.

SO7 Increase interest in soil health through a digital marketing strategy plan

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INNOVATIVE BUSINESS MODELS FOR SOIL HEALTH

https://novasoil-project.eu

